

Srijan Anand Gupta

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SUMMARY

Final-year Computer Science undergraduate (Data Science specialization) building end-to-end machine learning solutions — from classical algorithms implemented from scratch to rigorously evaluated pipelines (cross-validation, class-imbalance handling) and applied LLM/RAG systems. Comfortable across the full workflow: data preprocessing, model development, explainability, and deployment.

SKILLS

Languages: Python, SQL

Libraries & ML: Pytorch, Pandas, NumPy, Scikit-learn, XGBoost, Feature Engineering, Cross-Validation, Class Imbalance Handling, Exploratory Data Analysis (EDA), Statistical Analysis

AI / LLM: LangChain, Retrieval-Augmented Generation (RAG), Vector Embeddings, Hugging Face

Databases: MySQL

Visualization: Apache Superset, Matplotlib, Streamlit

Tools: Git, Jupyter Notebook

PROJECTS

UPI Fraud Risk Manager

2026

Python · XGBoost · Scikit-learn · Streamlit

- Built a fraud-detection pipeline on a 6.3M-transaction PaySim dataset using 5-fold stratified cross-validation with out-of-fold threshold selection to prevent test-set leakage.
- Achieved 99.33% recall and 88.65% precision (F2: 96.99%) on a held-out test set; handled severe class imbalance via `scale_pos_weight` after diagnosing an out-of-memory failure with SMOTE.
- Added business-cost sensitivity analysis and feature/permutation importance for explainability; deployed as a Streamlit app.

RepoReader

2026

Python · LangChain · Hugging Face · Streamlit

- Built a Retrieval-Augmented Generation (RAG) tool (CLI + Streamlit) that answers natural-language questions about any GitHub repository, grounded strictly in retrieved code/docs with source attribution.
- Designed a modular pipeline (loader, chunking, embedding, retrieval, generation) with a hallucination guardrail; parsed Jupyter notebooks cell-by-cell and embedded chunks locally with Sentence-Transformers.
- Diagnosed a dense-retrieval failure mode (relevant chunk ranked #11) and validated cross-encoder reranking as a fix (#11 → #1).

Machine Learning Algorithms from Scratch

2026

Python · NumPy · Pandas

- Implemented Linear, Multiple Linear, and Logistic Regression from scratch using vectorized NumPy operations.
- Built reusable Gradient Descent and Normal Equation implementations without relying on ML libraries for training.
- Validated coefficients and predictions against scikit-learn implementations on the Diabetes dataset.

EDUCATION

Bhagwan Parshuram Institute of Technology — B.Tech in Computer Science (Data Science)

CGPA: 8.04

CERTIFICATIONS

NIELIT Delhi Centre — Certificate Course in Data Science using Python (60 Hrs, Grade 'S'): Covered Fundamentals of Python, EDA with Pandas, Matplotlib, and ML Algorithms (Regression, Classification & Clustering).

NPTEL — Human Computer Interaction (Elite, Score: 91%): Covered usability engineering, cognitive principles, interaction design, and user-centered system design.